

## 22PH202 – Electromagnetism and Modern Physics\

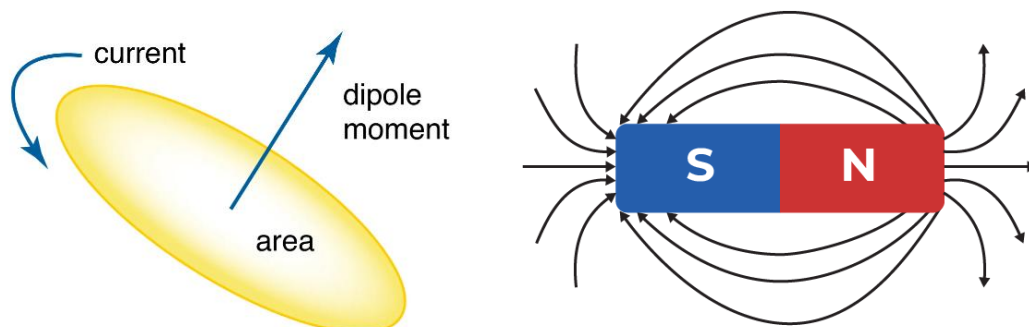
### Unit II – Magnetism

#### 2.3 Properties of magnet

##### 2.3.1 Magnetic Dipole and dipole moment

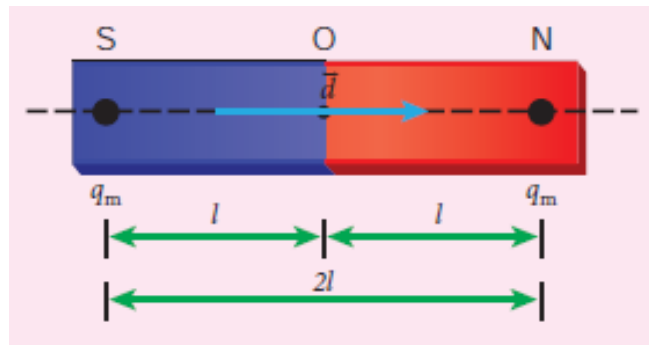
Magnetic dipoles are found to exist in magnetic materials (which are similar to electric dipoles in dielectric material). Magnetic dipoles may be thought of as small bar magnets composed of north and south poles instead of positive and negative electric charges.

Magnetic dipole, generally a tiny magnet of microscopic to subatomic dimensions, equivalent to a flow of electric charge around a loop. Electrons circulating around atomic nuclei, electrons spinning on their axes, and rotating positively charged atomic nuclei all are magnetic dipoles. The sum of these effects may cancel so that a given type of atom may not be a magnetic dipole. If they do not fully cancel, the atom is a permanent magnetic dipole, as are iron atoms.



**Figure 4. Magnetic dipole and poles of bar magnet**

**A magnetic moment** is a quantity that represents the magnetic strength and orientation of a magnet or any other object that produces a magnetic field. The magnetic dipole moment is defined as the product of its pole strength and magnetic length. It is a vector quantity, denoted by  $P_m$ . The SI unit of magnetic moment is  $A\ m^2$ .



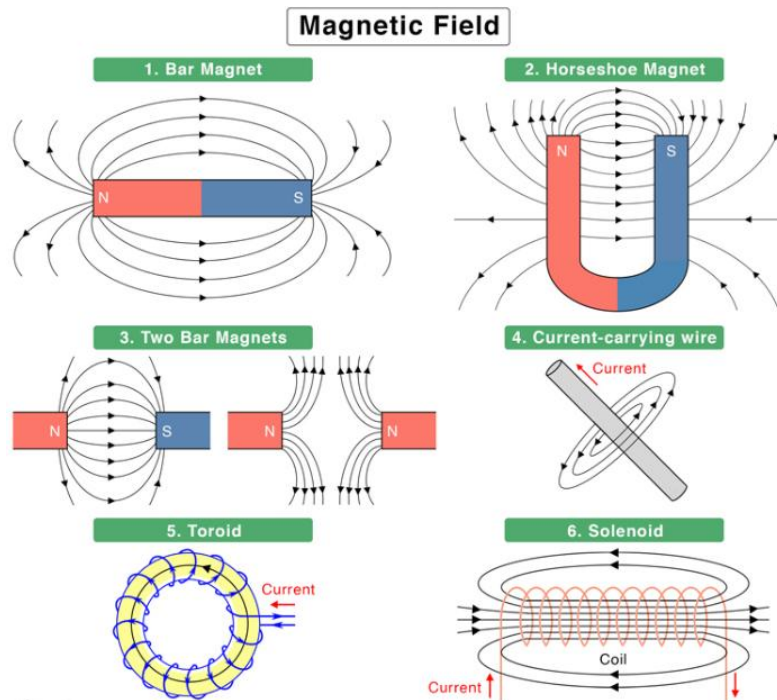
**Figure 5. Magnetic dipole moment**

$$P_m = 2lq_m \text{ (} q_m \text{: pole strength)}$$

### 2.3.2 Magnetic field and field lines

**Magnetic Field** is the region around a magnetic material or a moving electric charge within which the force of magnetism acts. Magnetic field is produced by permanent magnets and temporarily by electromagnets or superconducting magnets. **Magnetic field lines** are the lines in a magnetic field the tangent of which at any point will give the field direction at that point and its density gives the magnitude of the field.

- *Magnetic field lines are continuous closed curves. The direction of magnetic field lines is from North pole to South pole outside the magnet and from South pole to North pole inside the magnet.*
- *The direction of magnetic field at any point on the curve is known by drawing tangent to the magnetic field lines at that point.*
- *Magnetic field lines never intersect each other. Otherwise, the magnetic compass needle would point towards two different directions, which is not possible.*
- *The degree of closeness of the field lines determines the relative strength of the magnetic field. The magnetic field is strong where magnetic field lines crowd and weak where magnetic field lines are well separated.*



**Figure 6. Magnetic field and field lines of different magnets**

### 2.3.3 Magnetic flux and flux density

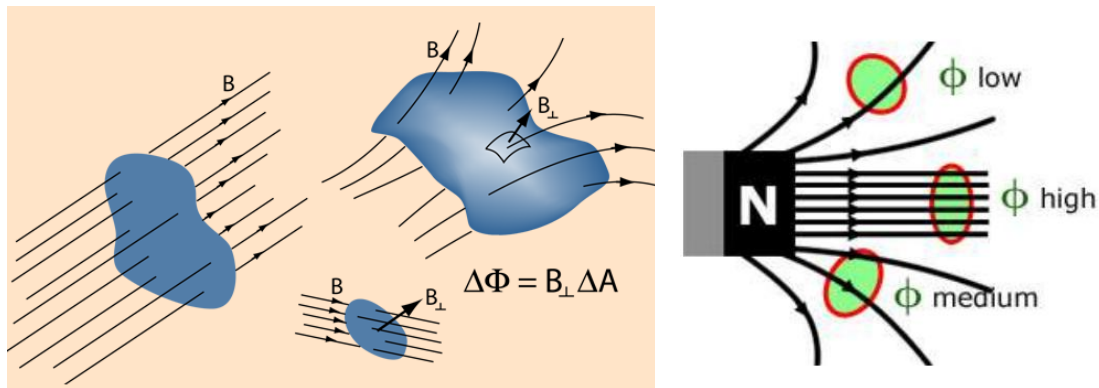
Magnetic flux is the product of the average magnetic field times the perpendicular area that it penetrates. It provides the measurement of the total magnetic field that passes through a given surface area. Here, the area under consideration can be of any size and under any orientation with respect to the direction of the magnetic field. Magnetic flux is a scalar quantity. The SI unit for magnetic flux is weber, which

is denoted by symbol Wb.

$$\Phi_B = \vec{B} \cdot \vec{A} = BA \cos \theta = B_{\perp} A$$

- When magnetic field (B) is normal to the surface i.e.,  $\theta = 0^\circ$ , the magnetic flux is maximum.
- When B is parallel to the surface i.e.,  $\theta = 90^\circ$ , the magnetic flux is 0.

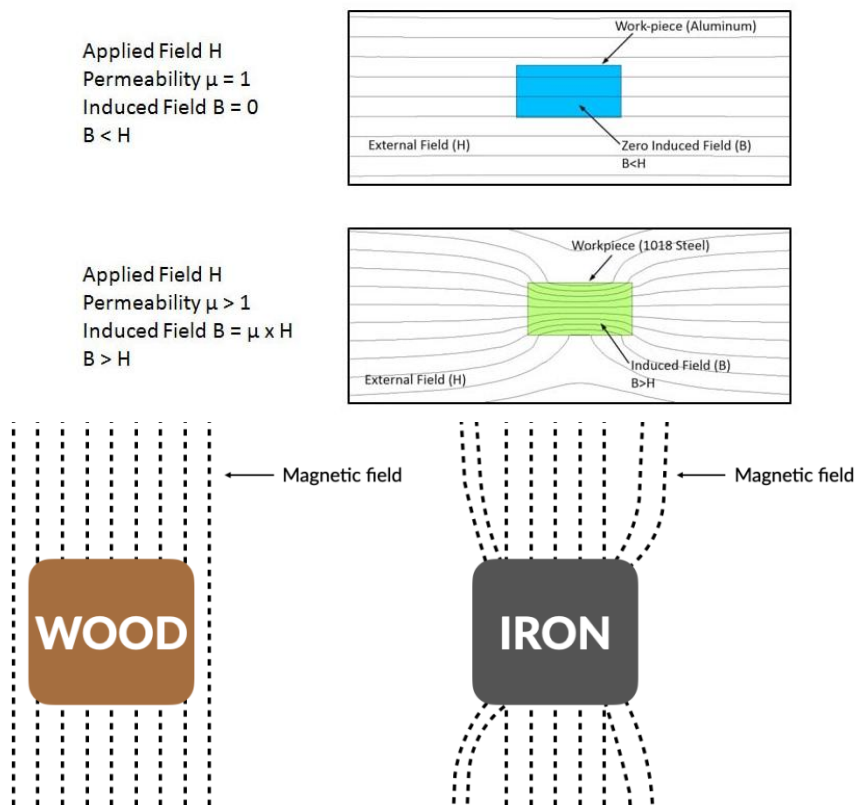
**Magnetic flux density** is defined as the force acting per unit current per unit length on a wire placed at right angles to the magnetic field.



**Figure 7. Magnetic flux**

### 2.3.4 Magnetic Permeability

Magnetic permeability is the characteristic of a material which represents the establishment of an induced internal magnetic field by an external magnetic field. The magnetic permeability is the proportionality between the induced field and the applied magnetic field strength.



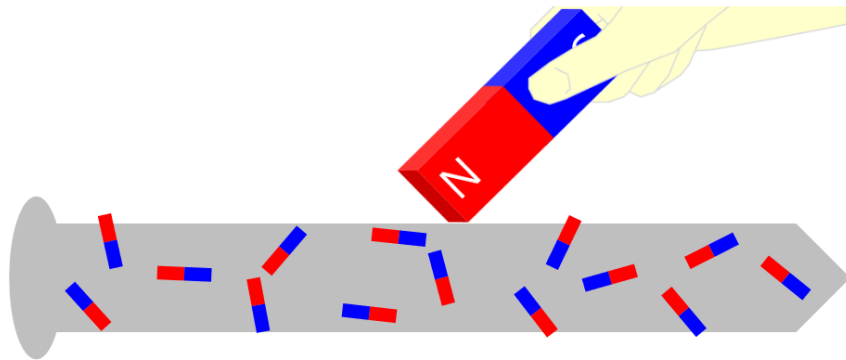
**Figure 8. Magnetic permeability of different materials**

### 2.3.5 Magnetic field strength

Magnetic field strength, the part of the magnetic field in a material that arises from an external current and is not intrinsic to the material itself. It is expressed as the vector  $H$  and is measured in units of amperes per metre.

### 2.2.6 Magnetization or Intensity of Magnetization

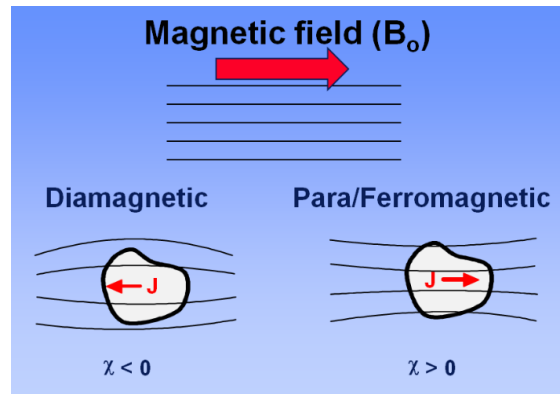
Magnetization is the density of magnetic dipole moments that are induced in a magnetic material when it is placed near a magnet. The magnetic effects of a material can also be induced by passing an electrical current through the material; the magnetic effect is caused by the motion of electrons in atoms, or the spin of the electrons or the nuclei.



**Figure 9. Magnetization of iron needle**

### 2.3.7 Magnetic susceptibility and Bohr magneton

In electromagnetism, the magnetic susceptibility is a measure of how much a material will become magnetized in an applied magnetic field. It is the ratio of magnetization  $M$  to the applied magnetizing field intensity  $H$ . This allows a simple classification, into two categories, of most materials' responses to an applied magnetic field: an alignment with the magnetic field,  $\chi > 0$ , called paramagnetism, or an alignment against the field,  $\chi < 0$ , called diamagnetism.



**Figure 10. Magnetic susceptibility of dia and para magnetic materials**

The orbital magnetic moment and the spin magneticmoment of an electron in an atom can be expressed interms of atomic unit of magnetic moment called as BohrMagneton

$$\mu_B = \beta = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2$$